1.2 How Are Cells Adapted to Their Functions?

Biology · Cell Structure and Organisation · Syllabus refs (G2 & G3): 1d

Name: _____ Class: _____ Date: _____

Learning outcome

- Explain how the structures of specialised cells are adapted to their functions (e.g. muscle cell: many mitochondria to supply more energy; root hair cell: large surface area of cell membrane for greater absorption; red blood cell: lack of nucleus allowing it to transport more oxygen).

Why are cells specialised?

In a large organism, different cells do different jobs. A **specialised cell** has a structure that is **adapted** (suited) to its function. Each one still has the usual cell parts, but some parts are changed, added or lost.

> **Key idea:** for every adaptation, say **what the feature is** and **how it helps the cell do its job**.

Differentiation

As an organism grows, its cells change to become **specialised**: each type of cell develops a structure that suits one particular job. This process is called **differentiation**.

Examples of specialised cells:

- **red blood cells** carry oxygen around the body;
- **nerve cells** carry messages between parts of the body;
- **intestinal cells** lining the gut absorb digested food;
- **root hair cells** in plants absorb water and mineral salts from the soil.

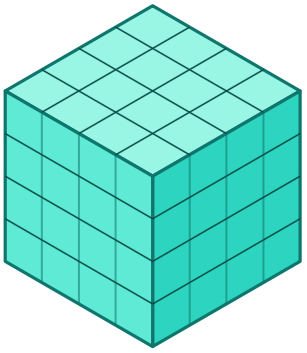
Below, you will look closely at red blood cells, muscle cells and root hair cells.

Surface area to volume ratio

Substances such as oxygen, water and mineral salts enter a cell through its **surface** (the cell membrane). The more surface a cell has for its size, the faster it can take substances in. We compare cells using the **surface area to volume ratio** (SA : V).

Try it: same volume, different shapes

Each shape is built from the same **64 small cubes** (volume = 64 units^3). Pick a shape and look at how much surface it has.



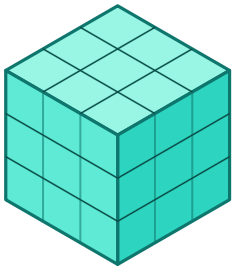
64	96	1.5 : 1
volume (units^3)	surface area (units^2)	SA : V ratio

Surface area = 2 × (4×4 + 4×4 + 4×4) = 96 units^2. SA : V = 96 : 64.

Same volume, but the **flatter** or **longer and thinner** the shape, the larger the surface area. That is why a red blood cell is a flattened disc and a root hair cell has a long, narrow extension.

Try it: cut a big cube into small cubes

Start with one 3 cm cube. Cut it into 1 cm cubes. The total **volume stays the same**, but look what happens to the surface.

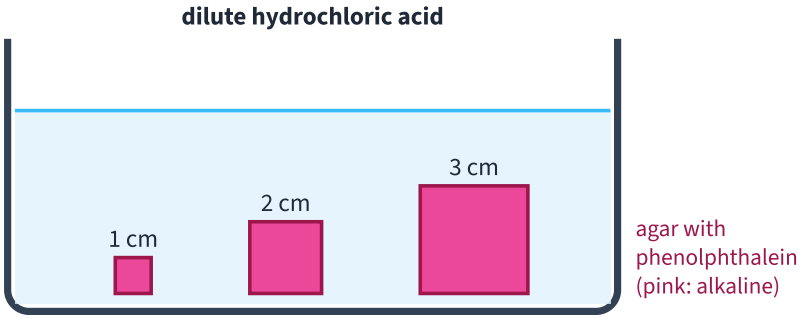


$27\ cm^3$	$54\ cm^2$	2 : 1
total volume	total surface area	SA : V ratio

One 3 cm cube: 6 faces × (3 × 3) cm^2 = 54 cm^2. Volume = 3 × 3 × 3 = 27 cm^3. SA : V = 54 : 27 = 2 : 1.

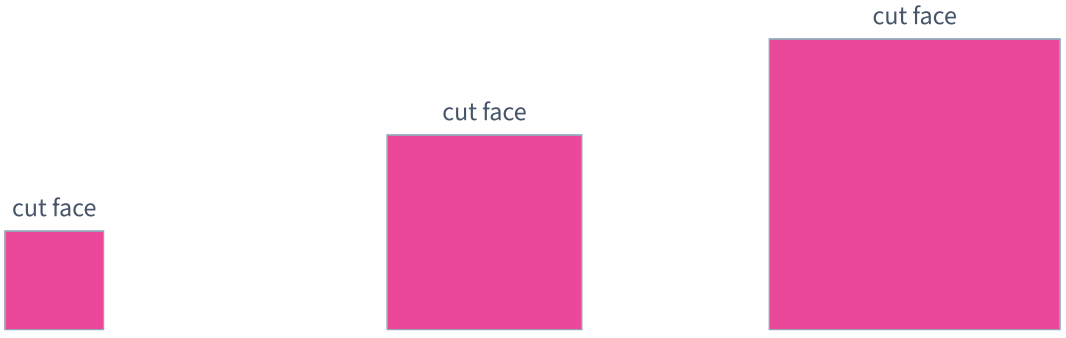
Link to cells: a large organism is made of many small cells rather than one huge cell. Each small cell has a large SA : V ratio, so substances can enter and leave it quickly enough.

Experiment: diffusion into agar cubes of different sizes



1. Make agar containing an alkali and **phenolphthalein** indicator. It is **pink** because it is alkaline.
2. Cut cubes of side **1 cm, 2 cm and 3 cm**.
3. Place them in **dilute hydrochloric acid**. As the acid **diffuses** into the agar, the indicator turns **colourless**.
4. Every few minutes, take the cubes out and **cut them in half** to see how far the acid has reached.

Run the experiment: press Play, or drag the time slider. You are looking at each cube cut in half: ■ still pink (no acid yet), □ colourless (acid has diffused in).



1 cm cube (SA : V = 6 : 1)

0% reached by acid

2 cm cube (SA : V = 3 : 1)

0% reached by acid

3 cm cube (SA : V = 2 : 1)

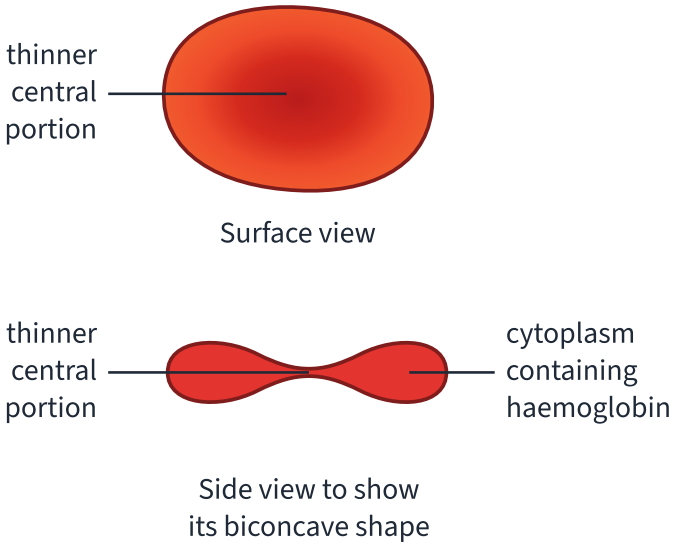
0% reached by acid

Cube side	Surface area	Volume	SA : V	Time to turn fully colourless*
1 cm	6 cm^2	1 cm^3	**6 : 1**	5 min
2 cm	24 cm^2	8 cm^3	**3 : 1**	10 min
3 cm	54 cm^2	27 cm^3	**2 : 1**	15 min

*In this simple model the acid moves 1 mm into the agar every minute from every face. Real times depend on the acid and the agar, and diffusion slows down as it goes deeper, but the pattern is the same.

> **Conclusion:** the **smallest** cube has the **largest SA : V ratio**, so the whole of its volume is reached by diffusion first. The bigger the cube, the longer the acid takes to reach its centre.

Red blood cell



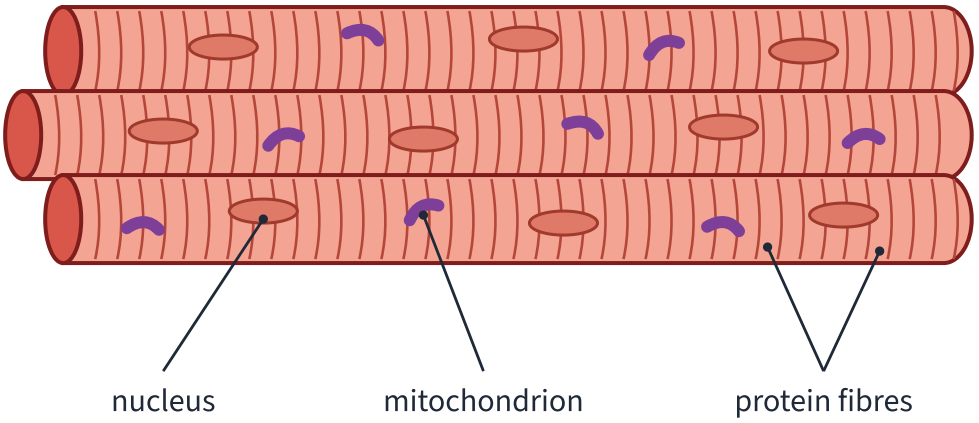
A red blood cell. Function: **transports oxygen** around the body.

Real red blood cells (scanning electron micrograph). Look for the biconcave discs: each one is dipped in the middle. Cells seen side-on look like thin bars.

Image: David Gregory & Debbie Marshall, CC BY 4.0, Wellcome Collection.

Adaptation	How it helps
Contains **haemoglobin**	Haemoglobin binds to oxygen, so the cell can carry oxygen around the body.
Circular, biconcave shape	Increases the surface area to volume ratio, so oxygen is taken in and released faster.
No nucleus	More space to pack in haemoglobin, so it can transport more oxygen.
Flexible	It can squeeze through narrow blood capillaries.

Muscle cell



A muscle cell. Function: **contracts and relaxes** to bring about movement.

Real muscle cells (light micrograph, stained pink). Each muscle cell is very long. The fine stripes across it come from the contractile protein fibres.

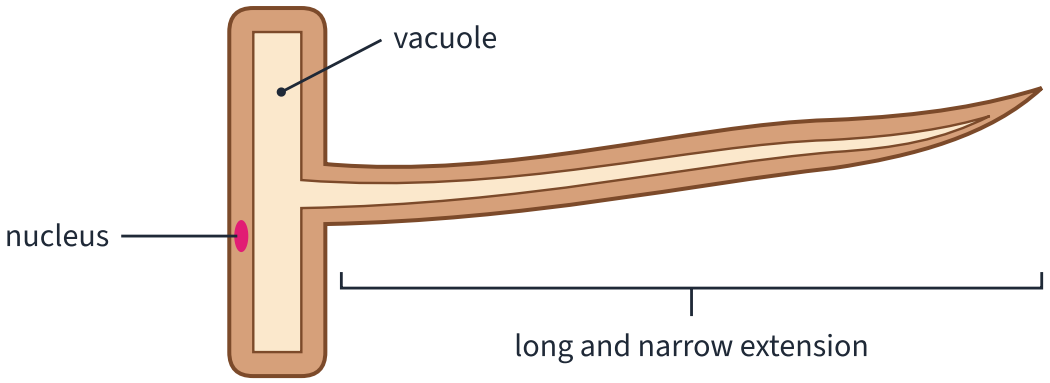
Image: Department of Histology, Jagiellonian University Medical College, CC BY-SA 3.0, via Wikimedia Commons.

Zooming in on heart muscle (illustration). The **mitochondria** (green) sit right next to the **myofibrils**, the contractile protein fibres, so the energy they release reaches the fibres quickly. Heart muscle never rests, so it is packed with mitochondria.

Image: BruceBlaus (Blausen Medical), CC BY-SA 4.0, via Wikimedia Commons.

Adaptation	How it helps
Contains **many mitochondria**	More aerobic respiration, so more energy is released for contraction.
Has **contractile protein fibres**	The fibres contract and relax to bring about movement.
Has **many nuclei**	Allows for cell division, growth and repair of the long cell.

Root hair cell



A root hair cell. Function: **absorbs water and mineral salts** from the soil.

A real root hair (light micrograph of a lily root). The long, thin extension on the left grows out of *one* cell on the surface of the root.

Image: Berkshire Community College Bioscience Image Library, CC0, via Wikimedia Commons.

Adaptation	How it helps
Long and narrow extension (the root hair)	Increases the surface area to volume ratio, so water and mineral salts are absorbed faster.
Cell sap in the vacuole has a **lower water potential** than the soil water	Water enters the root hair cell by **osmosis**. (You will learn about osmosis in Topic 2.)
No chloroplasts	It grows underground with no light, so chloroplasts would have no use.

Writing about adaptations

Link the **feature** to its **effect** and the **function**:

feature → so that… → which helps the cell to…

Example: "A red blood cell has **no nucleus**, **so** there is more space for haemoglobin, **so** it can transport more oxygen."

Check your understanding

1. Why does a red blood cell have no nucleus?

- ○ So it can divide faster
- ○ To make room for more haemoglobin
- ○ So it can absorb water
- ○ To release more energy

2. Muscle cells contain many mitochondria. What is the reason?

- ○ To make more proteins
- ○ To store cell sap
- ○ To release more energy by aerobic respiration for contraction
- ○ To absorb more light

3. How does the long extension of a root hair cell help the plant?

- ○ It gives the cell a fixed shape.
- ○ It increases the surface area to volume ratio for absorbing water and mineral salts.
- ○ It carries oxygen to the leaves.
- ○ It makes food by photosynthesis.

4. A cube-shaped cell has sides of 2 µm. Its surface area is 24 μm^2 and its volume is 8 μm^3. Work out the surface area to volume ratio (as a number : 1).

Your answer (: 1)

5. The biconcave shape of a red blood cell…

- ○ lets it squeeze through capillaries.
- ○ increases its surface area to volume ratio for faster uptake of oxygen.
- ○ gives it room for a nucleus.
- ○ makes it contract.